Trends in energy intake among US children by eating location and food source, 1977–2006

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Background Little is known about the impact of location of food consumption and preparation upon daily energy intake for children. Objective To examine trends in daily energy intake by children for foods eaten at home or away-from-home, by source of preparation, and for combined categories of eating location and food source. Subjects The analysis uses data from 29,217 children aged 2–18 years from the 1977–1978 Nationwide Food Consumption Survey, 1989–1991 and 1994–1998 Continuing Survey of Food Intake by Individuals, and 2003–2006 National Health and Nutrition Examination Surveys. Methods Nationally representative weighted percentages and means of daily energy intake by eating location were analyzed for trends from 1977 to 2006. Comparisons by food source were examined from 1994 to 2006. Analyses were repeated for 3 age groups: 2–6, 7–12, and 13–18 year olds. Difference testing was conducted using a t test. Results Increased energy intake (+179 kcal/d) by children from 1977–2006 was associated with a major increase in calories eaten away-from-home (+255 kcal/d). The percentage of kcal/d eaten away-from-home increased from 23.4% to 33.9% from 1977–2006. No further increase was observed from 1994–2006, but the sources of calories shifted. The percentage of calories from fast food increased to surpass intake from schools and become the largest contributor to foods prepared away-from-home for all age groups. For foods eaten away-from-home, the percentage of kcal/d from stores increased to become the largest source of calories eaten away-from-home. Fast food eaten at home and store-bought food eaten away-from-home increased significantly. Conclusion Eating location and food source significantly impact daily energy intake for children. Foods prepared away-from-home, including fast food eaten at home and store-prepared food eaten away-from-home, are fueling the increase in total calorie intake. However, further research using alternative data sources is necessary to verify that store-bought foods eaten away-from-home are increasingly store-prepared.